

D-D HELIUM-3 BREEDING SYSTEM

Two Auxiliary Cylindrical Breeder Cores - B1 and B2

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HANDLING	EYES ONLY - AUTHORIZED RECIPIENTS ONLY

1.0 - PURPOSE

This document defines the two auxiliary cylindrical D-D breeder cores supporting A.R.C.-6. These systems produce direct helium-3, generate managed tritium reserves, and provide isotope-processing support. They are not included in the six-primary-reactor A.R.C.-6 count.

2.0 - CONFIGURATION

Item	Certified configuration
Primary A.R.C.-6 reactors	6 parallel high-field fusion cores in the central hexagonal array
Auxiliary breeder cores	2 cylindrical pulsed D-D cores in separate shielded bays
Count rule	Only the six primary cores define A.R.C.-6; breeder cores B1 and B2 are auxiliary
Field relationship	Breeder systems are magnetically isolated and do not serve as A.R.C. Field anchors

3.0 - REACTION PATHWAYS

- $D + D \rightarrow He-3 + n + 3.27 \text{ MeV}$.
- $D + D \rightarrow T + p + 4.03 \text{ MeV}$.
- $T \rightarrow He-3 + \text{electron} + \text{antineutrino}$; half-life approximately 12.32 years.
- Lithium-6 blankets generate additional tritium for the strategic decay reserve; they do not create immediate helium-3.

4.0 - CYLINDRICAL BREEDER DESIGN

Each breeder core uses a cylindrical pulsed magnetized-target chamber with replaceable annular blanket cartridges, axial fuel injection, radial neutron shielding, and end-cap exhaust recovery. The cylindrical geometry differs deliberately from the six primary reactor modules and supports rapid remote replacement of isotope-production assemblies.

Parameter	Nominal certified range
Effective ion temperature	300-500 million K
Pulse duration	0.1-5 s
Process mode	Magnetized-target / compact-torus compression
Direct He-3 recovery	Exhaust and fuel-processing stream separation
Strategic decay reserve	20-30 year managed tritium inventory
Process pressure	kPa-to-MPa depending on stage
Vacuum-jacket pressure	Below 1×10^{-3} Pa normal

5.0 - CORE-SPECIFIC DUTIES

- Breeder Core B1: tritium breeding blanket operation, direct He-3 recovery, and plasma-facing material conditioning.
- Breeder Core B2: He-3 extraction, isotope separation, coolant conditioning, and reserve-inventory processing.
- Either breeder core may assume reduced-duty backup functions for the other, but neither substitutes for a primary power reactor.

6.0 - INTERFACE WITH A.R.C. FIELD

The breeder cores support fuel continuity but are not required to generate the A.R.C. coherence envelope. Their pulse schedules are phase-isolated from the six primary reactors to prevent magnetic and acoustic transients from disturbing the photonic lattice, alignment rings, or coherence spine.

7.0 - INVENTORY TIMELINE

The 2051 strategic inventory includes direct helium-3 recovered from D-D campaigns and tritium accumulated since precursor operations began in 2032. Managed aging provides meaningful tritium-to-helium-3 conversion while new production supports later decades.

8.0 - SAFETY

- Independent shielding, vacuum, cryogenic, and tritium boundaries for each breeder bay.
- Double-contained isotope piping and monitored hold-up storage.
- Automatic pulse inhibition during A.R.C. Field retuning or primary-array magnetic transients.
- No AGI command can bypass breeder-core hardwired protection or isotope accountancy.

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